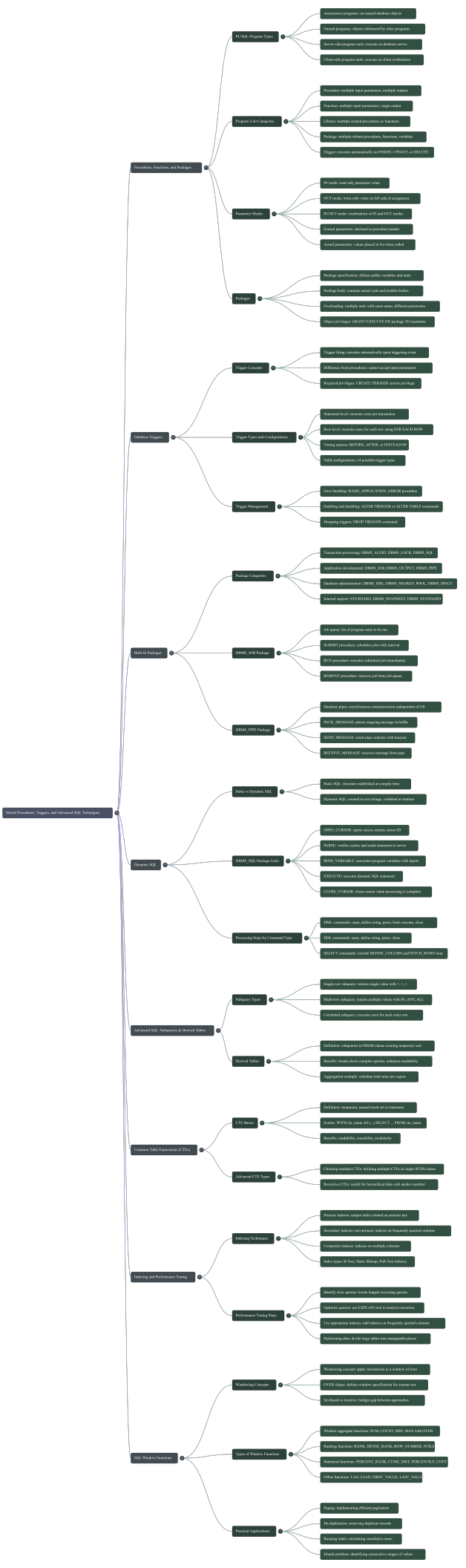

Stored Procedures, Triggers, and Advanced SQL Techniques
Procedures, Functions, and Packages
PL/SQL Program Types
Anonymous programs: un-named database objects
Named programs: objects referenced by other programs
Server-side program units: execute on database server
Client-side program units: execute on client workstation
Program Unit Categories
Procedure: multiple input parameters, multiple outputs
Function: multiple input parameters, single output
Library: multiple related procedures or functions
Package: multiple related procedures, functions, variables
Trigger: executes automatically on INSERT, UPDATE, or DELETE
Parameter Modes
IN mode: read-only parameter value
OUT mode: write-only value on left side of assignment
IN OUT mode: combination of IN and OUT modes
Formal parameters: declared in procedure header
Actual parameters: values passed in list when called
Packages
Package specification: defines public variables and units
Package body: contains actual code and module bodies
Overloading: multiple units with same name, different parameters
Object privileges: GRANT EXECUTE ON package TO username
Database Triggers
Trigger Concepts
Trigger firing: executes automatically upon triggering event
Difference from procedures: cannot accept input parameters
Required privileges: CREATE TRIGGER system privileges
Trigger Types and Configurations
Statement level: executes once per transaction
Row-level: executes once for each row using FOR EACH ROW
Timing options: BEFORE, AFTER, or INSTEAD OF
Valid configurations: 14 possible trigger types
Trigger Management
Error handling: RAISE_APPLICATION_ERROR procedure
Enabling and disabling: ALTER TRIGGER or ALTER TABLE commands
Dropping triggers: DROP TRIGGER command
Built-In Packages
Package Categories
Transaction processing: DBMS_ALERT, DBMS_LOCK, DBMS_SQL
Application development: DBMS_JOB, DBMS_OUTPUT, DBMS_PIPE
Database administration: DBMS_DDL, DBMS_SHARED_POOL, DBMS_SPACE
Internal support: STANDARD, DBMS_SNAPSHOT, DBMS_STANDARD
DBMS_JOB Package
Job queue: list of program units to be run
SUBMIT procedure: schedules jobs with interval
RUN procedure: executes submitted job immediately
REMOVE procedure: removes job from job queue
DBMS_PIPE Package
Database pipes: synchronous communication independent of OS
PACK_MESSAGE: places outgoing message in buffer
SEND_MESSAGE: sends pipe contents with timeout
RECEIVE_MESSAGE: receives message from pipe
Dynamic SQL
Static vs Dynamic SQL
Static SQL: structure established at compile time
Dynamic SQL: created as text strings, validated at runtime
DBMS_SQL Package Units
OPEN_CURSOR: opens cursor, returns cursor ID
PARSE: verifies syntax and sends statement to server
BIND_VARIABLE: associates program variables with inputs
EXECUTE: executes dynamic SQL statement
CLOSE_CURSOR: closes cursor when processing is complete
Processing Steps by Command Type
DML commands: open, define string, parse, bind, execute, close
DDL commands: open, define string, parse, close
SELECT commands: include DEFINE_COLUMN and FETCH_ROWS loop
Advanced SQL: Subqueries & Derived Tables
Subquery Types
Single-row subquery: returns single value with =, <, >
Multi-row subquery: returns multiple values with IN, ANY, ALL
Correlated subquery: executes once for each outer row
Derived Tables
Definition: subqueries in FROM clause creating temporary sets
Benefits: breaks down complex queries, enhances readability
Aggregation example: calculate total sales per region
Common Table Expressions (CTEs)
CTE Basics
Definition: temporary, named result set in statement
Syntax: WITH cte_name AS ( ) SELECT ... FROM cte_name
Benefits: readability, reusability, modularity
Advanced CTE Types
Chaining multiple CTEs: defining multiple CTEs in single WITH clause
Recursive CTEs: useful for hierarchical data with anchor member
Indexing and Performance Tuning
Indexing Techniques
Primary indexes: unique index created on primary key
Secondary indexes: non-primary indexes on frequently queried columns
Composite indexes: indexes on multiple columns
Index types: B-Tree, Hash, Bitmap, Full-Text indexes
Performance Tuning Steps
Identify slow queries: locate longest executing queries
Optimize queries: use EXPLAIN tool to analyze execution
Use appropriate indexes: add indexes on frequently queried columns
Partitioning data: divide large tables into manageable pieces
SQL Window Functions
Windowing Concepts
Windowing concept: apply calculations to a window of rows
OVER clause: defines window specification for current row
Set-based vs iterative: bridges gap between approaches
Types of Window Functions
Window aggregate functions: SUM, COUNT, MIN, MAX with OVER
Ranking functions: RANK, DENSE_RANK, ROW_NUMBER, NTILE
Statistical functions: PERCENT_RANK, CUME_DIST, PERCENTILE_CONT
Offset functions: LAG, LEAD, FIRST_VALUE, LAST_VALUE
Practical Applications
Paging: implementing efficient pagination
De-duplication: removing duplicate records
Running totals: calculating cumulative sums
Islands problem: identifying consecutive ranges of values